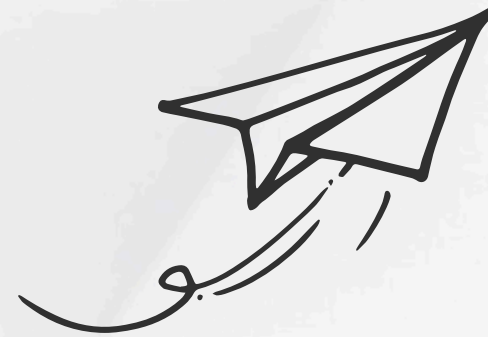


Node.js



RESTful API Design

What is REST?

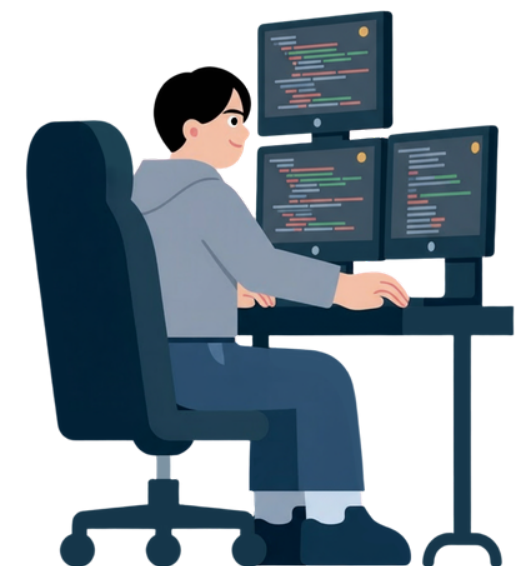
- REST = **Representational State Transfer**
- It is an architectural style for designing APIs.
- REST uses:
 - HTTP protocol
 - Resources (nouns)
 - Standard HTTP methods

HTTP Methods in REST

- **HTTP Methods :**

- **GET** : Retrieve data
- **POST** : Create data
- **PUT** : Update data
- **DELETE** : Remove data

```
GET      /products
POST     /products
PUT      /products/:id
DELETE   /products/:id
```



HTTP Status Codes

- **Important Status Codes**

- **200** : Success
- **201** : Created
- **400** : Bad Request
- **401** : Unauthorized
- **404** : Not Found
- **500** : Server Error



REST Example

- **Get All Products**

```
GET /products
```

- **Get one product**

```
GET /products/1
```

- **Create product**

```
POST /products
```

- **Update product**

```
PUT /products/1
```

- **Delete product**

```
DELETE /products/1
```

Database Integration

Why Do We Need a Database?

- Without database:
 - Data disappears when server restarts
 - No persistent storage
- With database:
 - Store users
 - Store products
 - Store orders
 - Real application behavior

Types of Databases

- SQL (Relational) stores data as Tables, rows, columns :
 - MySQL
 - PostgreSQL
 - SQL Server
- NoSQL : JSON-like documents
 - MongoDB
 - Firebase

Installing Database Driver

- Why **mysql2**?
 - Promise support
 - Modern driver
 - Works well with `async/await`

```
npm install mysql2
```

- **config/db.js**

```
const mysql = require("mysql2/promise");

const pool = mysql.createPool({
  host: "localhost",
  user: "root",
  password: "mineadub",
  database: "antecommerce",
});

module.exports = pool;
```

Using Environment Variables

- Install **.env**
- Create **.env**
- Load in **app.js**

```
npm install dotenv
```

```
DB_HOST=localhost  
DB_USER=root  
DB_PASSWORD=password  
DB_NAME=shop
```

```
require("dotenv").config();
```